SCD HUB · EXOTOPIA.ORG · ECO OPS PLATFORM

Eco Ops API — 8 Architectural Decisions

Discussion document for team review before Phase 1 build begins

MEETING / REVIEW DATE	ATTENDEES	VERSION / REF
		SPEC_ECO_OPS_API.md §8 · July 2026

> **What this document is for.** The Eco Ops API spec (SPEC_ECO_OPS_API.md) is ready to build against, but eight architectural questions need a team decision before work begins on Phase 1. Each question below states the issue, the options, and a staff recommendation. The goal of this session is to record a decision, a brief rationale, and an owner for each item. Unresolved questions will block specific Phase 1 work streams as noted.
>
> **Scope reminder.** Phase 1 covers: Supabase eco_ops schema, water quality data entry wizard, site map (Leaflet), country standards table, and the water-quality-alert Edge Function. Target jurisdictions: US, Kenya, Costa Rica, Canada, UK.

INFRASTRUCTURE

Q 1 Supabase project — shared or dedicated?

INFRASTRUCTURE

The blog commentary system (members, connections, comments) lives in the *public* schema of an existing Supabase project. The eco ops system needs its own *eco_ops* schema with eight new tables, new Row Level Security policies, and new Edge Functions. Should all of this live in the same project — sharing Auth users — or in a separate Supabase project with its own Auth?

If shared: a single login works across blog commentary and eco ops monitoring. Simpler for members who do both. One billing account. Schema separation keeps the tables clean.
If dedicated: clean isolation; Edge Function cold-start budgets are not shared; easier to hand off eco ops to a partner org later; requires a separate sign-in for users doing both.

STAFF RECOMMENDATION

Shared project with separate `eco_ops` schema. A single Auth user table is the right choice for participants who are both community commenters and field monitors. The eco ops schema provides adequate isolation. Revisit as a dedicated project only if the Row Level Security policies become too complex to maintain alongside the public schema's policies.

OPTIONS

- ☐ Shared project, separate `eco_ops` schema (recommended) *← recommended*
- ☐ Dedicated Supabase project — separate Auth, separate billing
- ☐ Other: ____________

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		Schema creation + Edge Function setup

Q 2 Map tile provider for Leaflet

INFRASTRUCTURE

The site map (EcoOpSiteListPage.vue) uses Leaflet, which needs a tile provider for the base map. Three realistic options vary in cost, attribution requirements, and offline viability. This matters for field use: monitors in Lamu or Costa Rica may be operating on intermittent 3G.

STAFF RECOMMENDATION

OpenStreetMap for Phase 1 — free, no API key, required attribution is a small footer credit. Add a Workbox service worker to cache tiles for the 20km radius around each registered site, making offline field use viable. Revisit MapLibre + self-hosted tiles in Phase 3 if tile-caching becomes complex or if we want custom styling.

OPTIONS

- ☐ OpenStreetMap — free, attribution required, offline-cacheable (recommended) *← recommended*
- ☐ MapLibre GL + self-hosted tiles (Protomaps / pmtiles) — free, more control, more setup
- ☐ Mapbox — paid (~$5/1,000 map loads), polished, easiest to style
- ☐ Other: ____________

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		EcoOpSiteListPage map implementation

DATA & STORAGE

Q 3 Photo storage — Supabase Storage only or IPFS archival?

DATA

Every monitoring record and knowledge record can include photos (bloom evidence, site conditions, drag-cloth ticks, elder knowledge sessions). Photos need to be stored, retrievable by access tier, and — for archival records — permanently accessible even if SCD Hub ceases operations.

Supabase Storage is built in and handles access tiers natively. IPFS (via Pinata) gives permanent content-addressed storage, which is already used for certificate SVGs. Adding IPFS for monitoring photos means any photo in an eco ops record can be independently verified and retrieved regardless of platform status — important for elder knowledge recordings and legally significant monitoring evidence (CSO events, PFAS results).

STAFF RECOMMENDATION

Supabase Storage as primary (fast, access-tier-aware). IPFS via Pinata for archival pin on any record flagged as: (a) elder knowledge / language documentation, (b) regulatory evidence (PFAS, CSO event), or (c) any photo the contributing community explicitly requests to archive. This is a per-record opt-in, not all-photos-always.

OPTIONS

- ☐ Supabase Storage primary + selective IPFS pin for archival records (recommended) *← recommended*
- ☐ Supabase Storage only — simpler, no IPFS cost or complexity
- ☐ IPFS-first (Pinata) for all photos — maximum permanence, higher cost, slower UX

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		MonitoringEntryPage photo upload step

Q 4 PFAS lab integration — manual entry or lab API?

DATA

PFAS sample results (EPA Method 533/537.1) are produced by certified labs. Currently the spec assumes manual data entry by the user after they receive their lab report. An alternative is to partner with Eurofins, TestAmerica, or a similar network — the lab posts results via API directly to the monitoring record, bypassing manual transcription and reducing error.

Lab API integration would require: a partnership agreement with ≥1 lab network, a chain-of-custody token linking our sample ID to the lab's accession number, and a webhook receiver Edge Function. It is not a Phase 1 blocker, but the schema needs to accommodate it from day one (the `pfas_compounds` jsonb column already does).

STAFF RECOMMENDATION

Manual entry for Phase 1 — the schema supports it. Reach out to Eurofins (US + EU + UK presence) and TestAmerica in parallel as partnership conversations, targeting Phase 3 integration. The key Phase 1 decision is whether to display a "submit to lab" button (providing a pre-filled chain-of-custody PDF) or to leave lab logistics entirely to the user.

OPTIONS

- [ ] Manual entry Phase 1 + pursue lab partnership for Phase 3 API integration (recommended) *← recommended*
- [ ] Manual entry only — no lab API ever; simpler but more error-prone
- [ ] Prioritise lab API partnership now — delay PFAS feature until integration is ready

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		PFAS entry form design only

USER EXPERIENCE & FIELD USE

Q 5 Mobile offline resilience for the monitoring wizard

UX / FIELD

Monitors in Lamu County, Mpeketoni, or a remote Costa Rica watershed may lose connectivity mid-entry. If the monitoring wizard (MonitoringEntryPage) does not persist its state locally, a dropped connection means lost data. The question is how much offline investment to make in Phase 1 versus deferring to Phase 4.

Minimum (sessionStorage): drafts survive accidental navigation away but not a closed tab or power loss. Low cost, one composable method.
Standard (localStorage + sync queue): survives closed tab; submitted data queues offline and syncs on reconnect. ~1 week of work.
Full PWA: service worker, background sync, offline map tiles, full offline-first experience. Phase 4 scope.

> **STAFF RECOMMENDATION**
>
> Standard (localStorage + sync queue) for Phase 1 — the Lamu use case demands it. The composable `useMonitoringWizard.ts` writes to `localStorage` on every step and queues failed submissions for retry. Full PWA tile caching deferred to Phase 4. Cost: approximately 3–5 extra days in Phase 1.

OPTIONS

- ☐ Minimum — sessionStorage only (fast to build, inadequate for field use)
- ☐ Standard — localStorage draft + offline submission queue (recommended) *← recommended*
- ☐ Full PWA now — complete offline-first; delays Phase 1 by ~3 weeks

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DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		MonitoringEntryPage wizard composable

Q 6 UK CSO outfall ID — manual lookup or EA database widget?

UX / FIELD

When a UK monitor records a CSO (combined sewer overflow) event, the monitoring form asks for the EA outfall permit number — a critical field for regulatory reporting. The EA's Event Duration Monitoring (EDM) database is publicly accessible. We can either ask the user to look it up themselves (friction, likely left blank) or build an auto-complete widget that queries the EDM API by postcode or GPS coordinates and returns nearby overflow points with their permit numbers.

STAFF RECOMMENDATION

Build the lookup widget. The EDM database has a public endpoint; the widget is a single composable function returning the nearest N outfall points for a coordinate. Without it, the `cso_outfall_id` field will be left blank by most users and the regulatory reporting value of the CSO record is halved. Cost: ~2 days. The UK launch story is significantly stronger with this included.

OPTIONS

- [ ] Manual text field — user looks up their outfall ID independently
- [ ] EA EDM lookup widget — auto-complete by coordinates or postcode (recommended) *← recommended*
- [ ] Skip UK CSO feature entirely for Phase 1 — address in Phase 3

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		UK water quality form + CSO Edge Function

LEGAL & POLICY

Q 7 Agency data submission — direct API or user-downloads-and-submits?

LEGAL

The `report-to-agency` Edge Function (§3.2) would send monitoring data to regulatory agencies: EPA Volunteer Monitoring Gateway (US), EA Data Returns API (UK), NEMA (Kenya), SENASA/SINAC (Costa Rica), provincial EAs (Canada). Submitting on behalf of users raises a question: if the data is inaccurate, does SCD Hub bear liability for a report filed with a regulator? Does the relevant agency even permit third-party submission?

Option (a) — direct API: maximum friction reduction, but SCD Hub is the named submitter in the agency system. Option (b) — export and self-submit: SCD Hub generates a formatted, ready-to-submit file; the user presses submit in the agency's own portal. Option (c) — MOU-first: pursue formal data sharing agreements with each agency before enabling any submission.

STAFF RECOMMENDATION

Option (b) for Phase 1 across all jurisdictions. Generate a formatted export (EPA-compatible CSV, EA XML, NEMA PDF) that the user submits themselves via the relevant agency portal. Zero liability exposure, same friction reduction for the user's preparation step. Pursue MOUs with EPA and EA in parallel; unlock direct API submission per-agency as agreements are in place. Kenya and Costa Rica portals may not have submission APIs — option (b) may be the permanent model there.

OPTIONS

- ☐ (a) Direct API submission — platform submits on user's behalf
- ☐ (b) Export + self-submit — platform generates the file, user submits it (recommended) *← recommended*
- ☐ (c) MOU-first — no submission feature until formal agency agreements are signed

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		report-to-agency Edge Function design

Q 8 SME engagement model — recognition only or paid advisory?

POLICY

Subject matter experts (limnologists, tick ecologists, PFAS specialists, language documentation practitioners) contribute peer review time to monitoring sites. The platform currently rewards them with: a Knowledge Contributor credential, attribution in the site library record, and — for multi-session contributors — a formal SME certification. No monetary payment pathway exists.

For professional SMEs giving substantive advisory time, recognition-only may be inadequate. A payment pathway would require: Stripe or PayPal for US/UK/Canada/CR; M-Pesa B2C (already built) for Kenya-based SMEs. It would also require `rate_per_hour` and `payment_status` columns in the `sme_engagements` table — easy to add in Phase 1 as nullable columns, impossible to add cleanly later without a migration.

STAFF RECOMMENDATION

Recognition-only for Phase 1. Add `rate_per_hour NUMERIC(10,2)` and `payment_status TEXT` as nullable columns to `sme_engagements` now — zero cost, preserves the Phase 3 option without a breaking migration. Revisit payment pathway design in Phase 2 planning once we have real SME engagement data showing how much time contributors are actually giving.

OPTIONS

- [ ] Recognition-only Phase 1 + nullable rate columns in schema for future (recommended) *← recommended*
- [ ] Recognition-only permanently — keep it a volunteer/community feature
- [ ] Build payment pathway in Phase 1 — Stripe + M-Pesa from the start

DECISION

DECISION OWNER	DUE / BY WHEN	BLOCKS PHASE 1 TASK
		sme_engagements table schema (column set)

GENERAL DISCUSSION NOTES